

RIDDHI GOSWAMI

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PROFILE

Computer Science student (CGPA: 9.98) awarded the UAE Golden Visa for academic excellence. Experienced in ML, AI, IoT, and software engineering. Proven leadership and technical delivery at Emirates Group, Jacobs, and ongoing internship at Esri. Skilled in software development and technical communication. Quick to grasp new technologies and concepts, with a strong ability to adapt and contribute in dynamic environments.

EDUCATION

BITS Pilani - Dubai Campus

B.E. (Hons) Computer Science: CGPA 9.98 - First Rank (2022-2024)

Dubai, UAE

Expected Aug 2025

The Indian High School

Grade 12 - Science Stream: Grade 98.6% - School Topper 2nd Position

Dubai, UAE

2021

The Indian High School

Grade 10: Grade 97%

Dubai, UAE

2019

EXPERIENCE

Software Developer Intern

Esri

Jan. 2025 – Present

Sharjah, UAE

- Contributed to the development and debugging of advanced geospatial tools in the Oriented Imagery (OI) division of ArcGIS Pro, working primarily in C++ with integration across .NET, WPF, and COM-based systems.
- Authored internal developer documentation including an intern onboarding guide (covering Devtopia setup, build systems, debugging, and DLL management) and technical workflow diagrams.
- Proposed future-facing testing strategies (CUIT and graphic harnesses) to improve automated UI validation.
- Resolved critical bugs affecting UI bindings, image overlays, viewer resets, and Geoprocessing (GP) tool crashes, with deep investigation into event dispatching, map-view state, and metadata parsing.
- Enhanced GP tool workflow to support video and mosaic ingestion into the OI system by extracting geospatial metadata and transforming into compliant OI dataset.
- Worked on a tool to export depth imagery layers as georeferenced TIFFs, aligning outputs with image metadata and camera footprints for use in 3D analysis and reconstruction.

Software Developer - Structure Team

Jacobs

Jun. 2024 – Aug. 2024

Dubai, UAE

- Engineered dynamic desktop applications using WPF with .NET 8.0 and integrated PyShiny for automating complex structural design workflows.
- Applied advanced NLP techniques—including Seq2Seq models, BERT, and fuzzy string matching—to standardise and enforce enterprise-wide naming conventions.
- Developed proficiency in Agile methodologies and Azure DevOps, leveraging design patterns and clean code principles for scalable and maintainable solutions.

Revenue Optimization Intern

Emirates Group HQ

Jun. 2023 – Aug. 2023

Dubai, UAE

- Developed machine learning models to accurately predict flight sell-out dates (MAE < 2 days) using high-volume datasets exceeding 500 million records.
- Engineered Big Data optimisations that significantly reduced ML model training time and improved pipeline efficiency.
- Designed interactive, data-rich dashboards with Plotly, Dash, and Streamlit to support real-time decision-making.
- Leveraged Snowflake for scalable data storage and fast analytical querying across distributed datasets.

MISCELLANEOUS EXPERIENCE

Management Lead, Google Developer Groups - BPDC
Professional Assistant - Computer Architecture, CS Dept. - BPDC
Event Manager, IEEE BPDC
Software Team Member, IFOR Drone and Robotics Research

Sept. 2024 – Present
Sept. 2024 – Jan. 2025
Sept. 2023 – Aug. 2024
Sept. 2023 – Aug. 2024

TECHNICAL SKILLS

Languages: C/C++, C#, Python, Java, HTML/CSS
Frameworks: .NET 8.0, React, Django, Flask, PyShiny, Dash, Streamlit
Data Processing: Pandas, Polars, MySQL, SnowflakeSQL, Excel
AI/ML: Sklearn, OpenCV, TensorFlow, Keras, Spacy, NLTK, LLMs
Misc. Tools: Git, Azure DevOps, Power BI, Arduino, ROS/ROS2

SELECTED PROJECTS

Dynamic Backdoor Attackers in Federated Learning

Jan. 2025 – Present

- * Conducted empirical research on the resilience of Federated Learning models against diverse backdoor attacks using CIFAR-10 and SimpleCNN in non-IID settings.
- * Identified the impact of trigger pattern consistency on attack stealth and effectiveness, contributing to insights on secure FL architecture design.
- * Currently authoring a journal paper based on this research for academic publication.

Automated Detection and Structural Health Monitoring

Nov. 2024 – Present

- * Designed a Deep Learning-based pipeline for automated structural health monitoring of civil infrastructure.
- * Trained custom ResNet, InceptionV3, and Transformer-based CoCa and RevCol models on a proprietary defect dataset for multi-class classification.
- * Currently authoring a journal paper based on this research for academic publication.

Voice-Controlled Mobile Bot

Jan. 2024 – Jan. 2025

- * NLP + ROS workflow using WhisperAPI, GPT-4o, ElevenLabs, Bark.
- * Real-time voice command recognition and robot mobility control.
- * Presented and published my work in IEEE CINS 2024 conference: [DOI: 10.1109/CINS.2024.10864443](https://doi.org/10.1109/CINS.2024.10864443)

AlphaDock: DNA-Protein Complex Predictor

Jan. 2024 – Jan. 2025

- * Developed a computational biology framework for predicting DNA-protein docking configurations using TF binding affinity datasets.
- * Analysed transcription factor interactions using UniProbe and CIS-BP databases for complex structure predictions.
- * Presented as a final-year research project under faculty review for evaluation and distinction.

CERTIFICATIONS AND ACCOLADES

Google x Kaggle Gen AI Intensive
HarvardX CS50 AI
NNs and Deep Learning - DeepLearning.AI
ACM Datathon 1st Place
ADU Datathon 2nd Place - WiDS
ML Project Structuring - DeepLearning.AI
Microsoft Learn AI Challenge
HarvardX CS50x Cert.
GDSC Ideathon Winner
Postman API Student Leader

Apr. 2025
Jul. 2024
Jul. 2024
Mar. 2024
Mar. 2024
Jan. 2024
Aug. 2023
Jan. 2023
Feb. 2023
2022–Present